MAT2040 Linear Algebra

Midterm Exam
SSE, CUHK(SZ)

25 June 2022

Student ID: _____

- Answer each question in the space after the question.
- Unless otherwise specified, be sure to give **full explanations** for your answers. The **correct reasoning** alone is worth **more credit** than the correct answer by itself.

This page has no questions

Question	1	2	3	4	5	6	7	8	Sum
Points									

Blank page

Question 1 ..... *10 points*

Please find all the Matrices $A \in \mathbb{R}^{2\times 2}$ which satisfy $A^2 = I_2$.

Question 2 . . . . . *20 points*

Solve the following linear systems

$$(1)\ \begin{cases} x_1 - x_2 + 2x_4 + x_5 = 0 \\ 3x_1 - 3x_2 + 7x_4 = 0 \\ x_1 - x_2 + 2x_3 + 3x_4 + 2x_5 = 0 \\ 2x_1 - 2x_2 + 2x_3 + 7x_4 - 3x_5 = 0 \end{cases}$$

$$(2)\ \begin{cases} x_1 + 2x_2 + 3x_3 = 8 \\ 2x_1 + 5x_2 + 9x_3 = 16 \\ 3x_1 - 4x_2 - 5x_3 = 32 \end{cases}$$

Question 3 ..... *20 points*

Compute the following matrix operations by using *matrix partition*:

(1) Matrix Multiplication $\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ -2 & 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 & 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{bmatrix}$

(2) Inverse Matrix of $\left[\begin{array}{cc|ccc} 2 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right]$

$= \left[\begin{array}{c|c} A & 0 \\ \hline 0 & B \end{array}\right] = \left[\begin{array}{c|c|c} A & 0 & I_2 \\ \hline 0 & B & 0 \\ \hline & 0 & I_3 \end{array}\right]$

$\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -1 \end{bmatrix} A = I_2$

$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & -2 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -\frac{1}{3} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad B = I_3$

$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & -3 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Question 4 ..... *10 points*

We know $ABA^T = 2BA^T + I$ and $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$. Please compute B.

$A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

$\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$

$ABA^T = 2BA^T + I$

$BA^T = 2B + A^{-1}$

$(AB)^T = 2B^T + A^{-1}$

$ABA^T = \begin{bmatrix} b_{11} & b_{12} & 2b_{12} + b_{13} \\ b_{21} + 2b_{31} & b_{22} + b_{32} & 2b_{22} + 4b_{32} + b_{33} + 2b_{33} \\ b_{31} & b_{32} & 2b_{32} + b_{33} \end{bmatrix}$

$AB = 2B + (A^{-1})^T$

$2BA^T = \begin{bmatrix} 2b_{11} & 2(2b_{13} + b_{12}) & 2b_{13} \\ 2b_{21} & 2(2b_{23} + b_{22}) & 2b_{23} \\ 2b_{31} & 2(2b_{33} + b_{32}) & 2b_{33} \end{bmatrix}$

$B = \begin{bmatrix} 1 & & \\ & 1 & \end{bmatrix}$

$(AB - 2B)A^T = I$

$AB - 2B = (A^T)^{-1}$

Question 5 ..... *10 points*

Matrix $A \in \mathbb{R}^{n \times n}$ is invertible, and matrix B is generated by interchanging the i-th and j-th rows of A. (1) Please check whether B is invertible? why? (2) and compute AB^{-1}.

Question 6 ..... *10 points*

Please compute the determinant of the following n by n matrix:

$$\begin{bmatrix} 2 & 1 & 0 & \cdots & 0 & 0 \\ 1 & 2 & 1 & \cdots & 0 & 0 \\ 0 & 1 & 2 & \cdots & 0 & 0 \\ \cdots & & & & & \\ 0 & 0 & 0 & \cdots & 2 & 1 \\ 0 & 0 & 0 & \cdots & 1 & 2 \end{bmatrix}$$

Question 7 ..... *10 points*

Please compute the determinant of the following n by n matrix:

$$\begin{bmatrix} 1 & 2 & 2 & \cdots & 2 & 2 \\ 2 & 2 & 2 & \cdots & 2 & 2 \\ 2 & 2 & 3 & \cdots & 2 & 2 \\ \cdots & & & & & \\ 2 & 2 & 2 & \cdots & n-1 & 2 \\ 2 & 2 & 2 & \cdots & 2 & n \end{bmatrix}$$

Question 8 ..... *10 points*

Please compute the determinant of the following 4 by 4 matrix:

$$\begin{bmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ 1+a_2b_1 & 1+a_2b_2 & 1+a_2b_3 & 1+a_2b_4 \\ 1+a_3b_1 & 1+a_3b_2 & 1+a_3b_3 & 1+a_3b_4 \\ 1+a_4b_1 & 1+a_4b_2 & 1+a_4b_3 & 1+a_4b_4 \end{bmatrix}$$